# Algunas exploraciones en electorencefalografía cuantitativa

## De señal a biomarcador

Oscar Yáñez Suárez
oyanez@izt.uam.mx

Ingeniería Eléctrica / UAM Iztapalapa

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LINI
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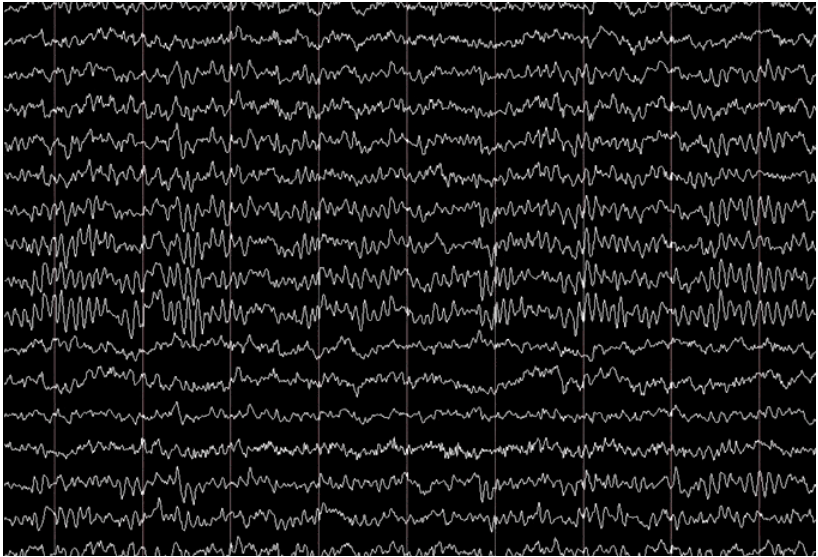
# Contenido

1. Motivación y Contexto
2. Análisis del EEG
3. Aplicaciones clínicas
4. Limitaciones y perspectivas

# Electroencefalografía



# ¿Por qué EEG cuantitativo?

- El análisis *clásico* del EEG parte de la inspección visual — subjetiva, lenta, dependiente del experto
- El qEEG extrae rasgos numéricos objetivos y reproducibles a partir de las señales crudas
- Posibilita los estudios poblacionales de gran escala, las interfaces cerebro-computadora en tiempo real, y el descubrimiento de biomarcadores

Confluencia de:

- Sistemas de alta densidad (256+ electrodos)
- Pipelines extendibles de software libre (MNE, BCI2000)
- Herramientas modernas de aprendizaje maquinal

## Referencias clave

Nuwer *et al.* (1998) establecen estándares básicos de qEEG [5]

Makeig *et al.* (2004) consolidan el análisis de la dinámica cerebral [6].

# EEG 101

## Generación

- Potenciales post-sinápticos en las dendritas apicales de neuronas piramidales
- *¡También potenciales de acción!*
- Conducción por el volumen de LCR/cráneo/piel cabelluda

## Interpretación *clásica* en bandas

| Banda | Rango (Hz) | Correlato cognitivo |
|---|---|---|
| $\delta$ | 0.5–4 | Sueño profundo, patología |
| $\theta$ | 4–8 | Memoria, somnolencia |
| $\alpha$ | 8–13 | Vigilia |
| $\beta$ | 13–30 | Cognición activa |
| $\gamma$ | 30–100+ | Fijación, percepción |

## Referencia clave

Fernandez-Ruiz *et al.* (2023) asocian las oscilaciones gamma con la operación de circuitos neuronales [8].

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# EEG 101

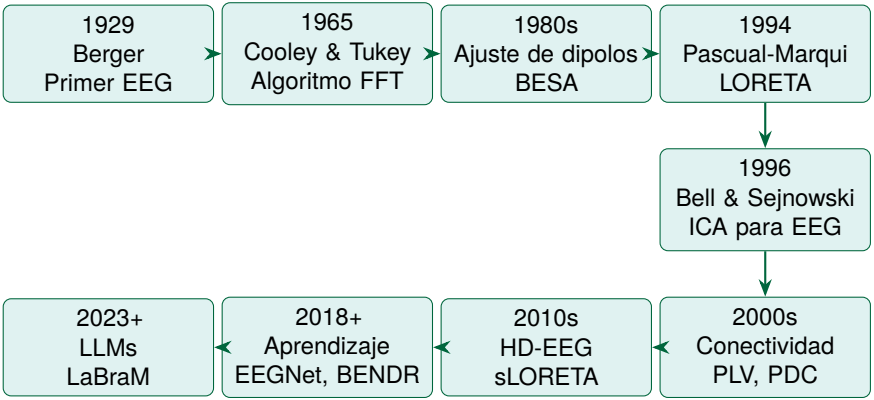
**Artefactos de medición**

- *Ocular*: parpadeos (EOG), movimientos sacádicos
- *Muscular*: EMG (>30 Hz)
- *Cardiaco*: ECG, BCG
- *Entorno*: ruido de linea, deriva electródica

**Nota práctica**

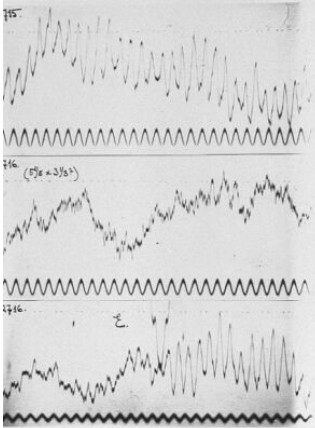
La elección de referencia (CAR, mastoides ligados, REST) impacta críticamente los resultados espectrales y de conectividad [7].

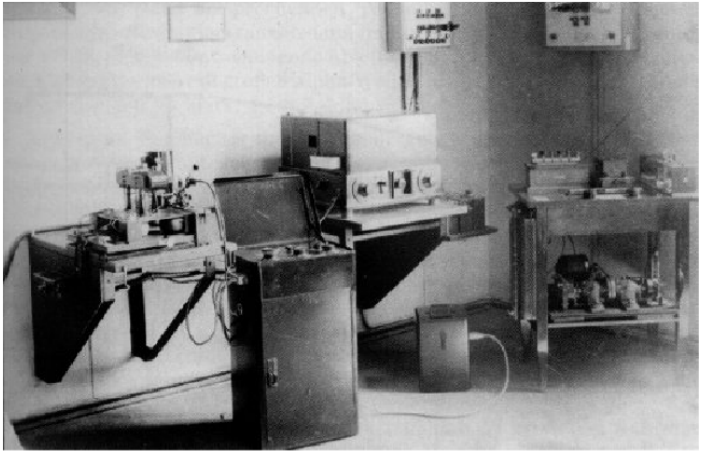
# Línea del tiempo de la qEEG




Referencias: [1, 2, 4, 3, 13]

# Hans Berger, 1929



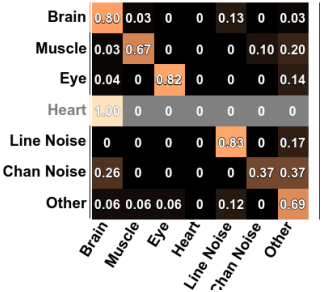


# Flujo de trabajo de preprocesamiento en qEEG

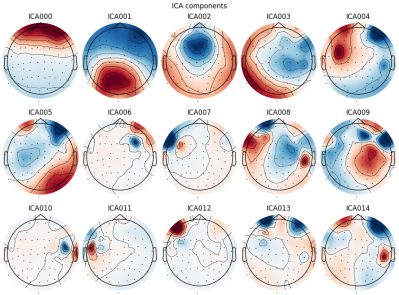
**Procesos comunes**

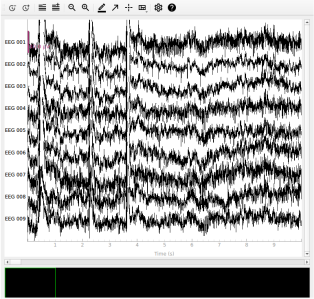
1. Cambio de referencia (CAR / REST)
2. Filtrado pasa/rechaza banda
3. Segmentación basada en tiempo o en marcas
4. Rechazo/interpolación de épocas
5. Artifact subspace reconstruction (ASR) [19]
6. Descomposición y clasificación de componentes ICA (ICLabel) [20]

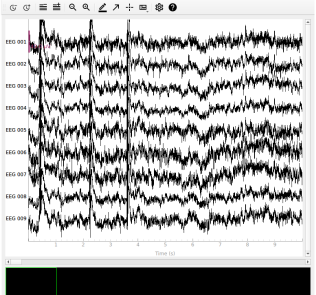
## Matriz de confusión ICLabel



# Reconstrucción post-ICLabel



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# Más allá de la TDF

**Descomposiciones ortogonales**

- CSP - Filtrado basado en componentes principales

**Descomposiciones redundantes**

- Descomposición empírica en modos (EMD)

**Representaciones tiempo-frecuencia**

- STFT / espectrograma
- BPts [**?**]
- Wavelets [11, 12]

# Functional & Effective Connectivity

**Undirected (functional) measures**

- **Coherence**: spectral correlation — volume-conduction sensitive
- **PLV** (phase-locking value): phase synchrony, ignores amplitude [13]
- **PLI / wPLI**: imaginary component only $\Rightarrow$ robust to zero-lag artefacts [14]
- **AEC** (amplitude envelope correlation): slow fluctuations in broadband power

**Directed (effective) measures**

- **Granger causality** / MVAR models [15]
- **DTF / PDC**: partial directed coherence in frequency domain
- **Transfer entropy**: model-free, information-theoretic
- **DCM**: Bayesian, mechanistic [16]

## Phase–amplitude coupling (PAC)

Coupling between $\theta$ phase and $\gamma$ amplitude tracks working memory load [17].

# Solving the EEG Inverse Problem

**The inverse problem is ill-posed**

- Infinitely many source configurations can produce the same scalp topography
- Regularisation and anatomical priors are essential

**LORETA family** [4]

- **LORETA**: Laplacian smoothness prior
- **sLORETA**: zero localisation error for single dipoles
- **eLORETA**: exact, standardised; gold standard for resting-state

**Beamforming**

- **LCMV**: scalar/vector; optimal for ERPs
- **DICS**: frequency-domain; ideal for oscillatory sources [18]
- Requires accurate covariance — sensitive to artifact residuals

**MRI-constrained methods**

- Individual head models (BEM, FEM)
- Cortical surface constraint (FreeSurfer parcellations)
- Combined EEG-fMRI asymptotic approaches

# Machine & Deep Learning on EEG

## Classical feature engineering

- Band power, spectral entropy, Hjorth parameters
- AR coefficients, coherence matrices
- Riemannian geometry of covariance matrices [23]
- CSP + LDA pipeline (motor imagery BCI)

## Shallow classifiers

- SVM, Random Forest, LDA
- XGBoost on spectral features
- Transfer learning via Euclidean alignment

## Deep learning architectures

- **EEGNet**: compact depthwise CNN; cross-task generalisation [24]
- **ShallowConvNet / DeepConvNet** [25]
- **EEG-Conformer**: self-attention over channels & time [26]
- **BENDR**: contrastive self-supervised pretraining [27]
- **LaBraM**: foundation model, cross-dataset pretraining [28]

## Open challenge

Cross-subject and cross-device generalisation remain unsolved without domain adaptation.

# EEG Microstates & Nonlinear Complexity

## EEG Microstates

- Global scalp topography quasi-stable for ∼60–120 ms
- Canonical 4–7 classes (A–G) via k-means / AAHC [29]
- Sequence statistics: duration, occurrence rate, transition matrix
- Classes A–D map to dorsal attention, visual, somatomotor, DMN networks [30]
- Altered in schizophrenia, depression, Alzheimer's disease

## Nonlinear complexity measures

- **SampEn**: regularity of temporal patterns; reduced in coma [31]
- **Lempel–Ziv**: compressibility of binarised signal
- **DFA**: long-range temporal correlations
- **Lyapunov exponent**: phase-space divergence

## Brain criticality

Healthy resting-state EEG sits near a critical point; complexity shifts towards subcriticality in anaesthesia and DOC [32].

# Clinical Applications I: Epilepsy & Sleep

## Epilepsy

- **Automated seizure detection**: CNN/LSTM on raw iEEG reaches $>95\,\%$ sensitivity [33]
- **Seizure prediction**: phase-space features, 30–60 min pre-ictal window
- **Spike detection**: template matching, iEEG atlas
- HFOs (high-frequency oscillations) as epileptogenic zone markers

## Sleep staging

- AASM rules (W/N1/N2/N3/REM) automatable at $\sim 85\,\%$ epoch accuracy
- **U-Sleep**: universal model across PSG montages [34]
- Spindle & K-complex detection: matched filters + neural networks
- HRV–EEG coupling for sleep quality metrics

## Key toolboxes

EEGLAB, MNE-Python, FieldTrip, Brainstorm, YASA (sleep)

# Clinical Applications II: Psychiatry & Consciousness

## Psychiatric biomarkers

- **Depression**: frontal $\alpha$ asymmetry (F4-F3) predicts treatment response [35]
- **Schizophrenia**: MMN amplitude reduction; $\gamma$ oscillation deficit
- **ADHD**: elevated $\theta/\beta$ ratio (controversial as standalone biomarker)
- **Alzheimer's**: slowing of peak $\alpha$ frequency, reduced coherence

## Disorders of consciousness (DOC)

- **PCI** (perturbational complexity index, TMS-EEG): discriminates VS from MCS [36]
- Covert awareness detection via P300 BCI
- Resting-state connectivity predicts prognosis

## Neurofeedback

- Real-time SMR / $\theta$-$\beta$ training for ADHD
- SCP neurofeedback for ALS communication
- Closed-loop brain stimulation [37]

# Limitations & Open Problems

## Methodological challenges

- **Spatial resolution**: scalp EEG limited to ∼5–10 cm; MEG and iEEG superior
- **Reference problem**: no truly neutral reference exists [7]
- **Non-stationarity**: standard spectral assumptions often violated
- **Small samples**: most qEEG studies N $< 50$; risk of inflated effect sizes
- **Reproducibility**: preprocessing choices create a ”garden of forking paths”

## Deep learning–specific issues

- **Interpretability**: limited mechanistic insight from black-box models
- **Cross-subject generalisation**: large inter-individual variability
- **Cross-device transfer**: amplifier, electrode type, sampling rate differ across labs
- **Data scarcity**: clinical EEG rarely public (privacy)

## Replication crisis

Several qEEG biomarkers ($\theta/\beta$ ratio, $\alpha$ asymmetry) have failed independent replication at scale.

# Conclusions & Future Directions

## Summary

- qEEG has matured into a rich, multi-method discipline
- Automated preprocessing (ASR, ICLabel) has largely solved artifact rejection
- Connectivity and source imaging provide brain-network perspectives
- Deep learning achieves state-of-the-art on classification benchmarks
- Foundation models (LaBraM) suggest a paradigm shift ahead

## Open frontiers

- **Multimodal**: simultaneous EEG-fMRI-fNIRS
- **Wearable / ambulatory** HD-EEG (dry electrodes, OPM-MEG)
- **Causal biomarkers**: TMS-EEG perturbational approaches
- **LLMs + EEG**: decoding imagined speech
- **Federated learning** for privacy-preserving multi-site models

*"The brain is a noisy, non-stationary, high-dimensional system — and that is exactly what makes it interesting."*

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# ¡Gracias por su atención!

¿Preguntas?

*oyanez@izt.uam.mx*

Presentación y fuentes: `https://github.com/oscarys/2026_PIB`